ENGINEERING HANDOFF DOCUMENT · FEB 2026

Suitcase & Life®

A minimal, Apple-like editorial display shelf for a curated personal archive.

"We are Ankeet and Sruthi. We love to collect things. This is a sneak peek into our small archive of gathered curiosities." — @suitcaseandlife

Stack	React 19 (CRA) + FastAPI 0.110 + MongoDB 7
Data source	Public Google Sheet (CSV export, 5-min server cache)
Deployment	Emergent platform (Kubernetes / supervisor)
Live preview	`https://suitcase-archive.preview.emergentagent.com`
Repository	Use the platform's *Save to GitHub* action to push the codebase
Auth	None — public read-only site, anonymous "Add to Shelf" likes via localStorage

How to use this document. Hand it to an engineer or AI agent. It contains the full product brief, architecture, every API contract, every file's responsibility, the design system, the known quirks, and the prioritised backlog. Pair it with the GitHub repo (push via *Save to GitHub*) and they can pick up immediately.

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2. Product Overview & Personas

The product

A single-page editorial website that acts as a public "display shelf" for the curators' personal collection of objects (electronics, furniture, ceramics, cameras, etc.). The data lives entirely in a Google Sheet so the curators never have to touch code to add a new item. The site auto-syncs every 5 min.

Personas

Persona	Job-to-be-done	Key surface
Curators (Ankeet & Sruthi)	Add an object: photos, story, price, source, tags. Want zero friction. Never want to deploy code to add an item.	The Google Sheet (single source of truth)
Visitors (Instagram followers, friends, design lovers)	Browse the shelf, filter by category, read the story behind each piece, mark "Add to Shelf" for things they love.	Home (Hero → Bento → Archive), product modal, /my-shelf

Three sections (Home page)

1. **Cinematic Hero** with a blurred lifestyle background, oversized brand wordmark, and updated-weekly meta.
2. **Bento Grid** with 4 featured tiles + 1 mandatory black "Archive Note" accent card. Featured items are pulled from sheet flag; if <4 featured, top-liked items fill the rest.
3. **Full Archive** — sticky category filter pills, then a responsive grid of `ProductCard`s. 2 cols on mobile, 3 on tablet, 4 on desktop.

3. Feature Inventory

Implemented (fully tested)

- DONE Google Sheet ingestion: tolerant CSV parser, header normalisation, 5-min cache, manual `/api/refresh`.
- DONE 10-product sample fallback when the sheet is empty/unreachable.
- DONE Cinematic Hero with `EB Garamond` display + animated meta strip.
- DONE Bento Grid (responsive, 2-col on mobile).
- DONE Archive with sticky category filter pills.
- DONE Product Detail modal — separate desktop & mobile layouts:
  - Desktop: 7/5 grid, image left, fixed header + scrollable description + fixed action bar on the right.
  - Mobile: full-screen, single scrollable area, sticky close (top-right) and sticky bottom action bar (Add to Shelf + open link).
  - Swipe-left/right on mobile to change image.
  - Prev/next product navigation (keyboard ← →, on-screen chevrons).
- DONE Custom **scroll-lock**: page is frozen at the moment-of-click position; closing the modal restores that exact scrollY.
- DONE Custom backdrop overlay (z-99) that covers the navigation bar so the modal feels properly layered.
- DONE "Add to Shelf" button (renamed from "I'm interested"). Server counter + localStorage idempotence.
- DONE "My Shelf" page (renamed from "My List") at `/my-shelf` with a custom inline-SVG empty state (line-art bookshelf + confused dog).
- DONE About page with story, gallery, links.
- DONE Mobile responsive across all breakpoints; hamburger menu in the header.
- DONE Backend automated tests (10/10 pass), Frontend Playwright tests (100% on critical paths).

Planned / backlog

- P1 Header "Refresh" pill that calls `/api/refresh` for instant Sheet sync.
- P1 Open-Graph & Twitter card metadata per product (rich link previews).
- P1 Share-to-Instagram / copy-link button per product.
- P2 Lenis momentum scrolling.
- P2 View-transition / shared-element animation card → modal.
- P2 Keyboard arrow navigation inside the modal gallery.
- P3 Email capture: "Tell me when new things are added".
- P3 Server-side image proxy + responsive `srcset`.
- P3 "Share My Shelf" — encode liked-IDs into a URL hash for sharable visitor shelves.

4. Architecture & Code Map

```
/app
 ├─── backend/
```

```
│   ├── server.py              ← FastAPI app, Sheet fetch + cache, likes
│   ├── requirements.txt
│   ├── tests/                 ← pytest suite (10/10 passing)
│   └── .env                   ← MONGO_URL, DB_NAME, GOOGLE_SHEET_ID, etc.
├── frontend/
│   ├── public/
│   │   └── index.html         ← Custom Google Fonts links
│   ├── src/
│   │   ├── App.js             ← Routes, scroll-lock, like-set, ProductDetail
│   │   ├── App.css
│   │   ├── index.css          ← Global tokens, font-faces, smooth-scroll
│   │   ├── lib/
│   │   │   └── api.js         ← Axios calls + localStorage helpers
│   │   ├── components/
│   │   │   ├── Header.jsx     ← Fixed nav, mobile menu, time strip
│   │   │   ├── Hero.jsx       ← Cinematic blurred hero, brand wordmark
│   │   │   ├── BentoGrid.jsx  ← 4 featured + accent card
│   │   │   ├── Archive.jsx    ← Sticky filter pills + grid
│   │   │   ├── ProductCard.jsx
│   │   │   ├── ProductDetail.jsx ← Dialog (desktop + mobile layouts, swipe)
│   │   │   ├── Footer.jsx
│   │   │   └── ui/            ← shadcn components
│   │   └── pages/
│   │       ├── Home.jsx       ← Hero + Bento + Archive
│   │       ├── MyList.jsx     ← /my-shelf — empty SVG state + grid
│   │       └── About.jsx
│   └── .env                   ← REACT_APP_BACKEND_URL
├── memory/
│   ├── PRD.md                 ← Living product spec
│   └── test_credentials.md
├── test_reports/              ← iteration_*.json from the testing agent
└── design_guidelines.json
```

Service layout (managed by `supervisor`):

- **Backend** on `0.0.0.0:8001`, all routes prefixed with `/api`.
- **Frontend** on `:3000` (CRA dev server, hot-reload).
- **MongoDB** local, accessed via `MONGO_URL` env var.
- Kubernetes ingress routes `/api/*` to backend, everything else to frontend.

5. Backend API Reference

Method	Path	Purpose	Response shape
GET	`/api/products`	All products from the cached Sheet (or sample fallback) merged with MongoDB like counts.	`Product[]` (see §6)
GET	`/api/categories`	Distinct categories present in the sheet, alphabetised.	`string[]`

POST	`/api/products/{slug}/like`	Increment the visitor "Add to Shelf" counter in MongoDB.	`{ id: string, likes: number }`
GET	`/api/refresh`	Force-clear the Sheet cache and re-fetch immediately.	`{ ok: true, count: number }`

All endpoints are public (no auth). CORS allowed from `CORS_ORIGINS` env var.

6. Data Contracts

Google Sheet (source of truth)

The sheet must be shared as *"Anyone with the link → Viewer"*. Header row required, but **any casing/order** is accepted thanks to header normalisation.

Column	Required	Notes
`Name`	Yes	Used for the slug (`id`) too.
`Category`	Yes	Powers filter pills.
`Price`	No	Display string; aliases `Cost` accepted.
`Link`	No	Source URL — the open-link arrow in the modal.
`Media URLs`	Yes	4–6 URLs, comma or newline separated. `.mp4/.webm/.mov/.m4v` rendered as `<video>`.
`Description`	Recommended	The "story" shown in the modal.
`Store Name`	No	e.g. "Nordgreen Flagship".
`City`	No	e.g. "Copenhagen".
`Date Acquired`	No	Free-form string (display as-is).
`Featured`	No	`yes/no`, `true/false`, or `1/0`. 4 featured items surface in the Bento.
`Tags`	No	Comma-separated.

MongoDB collection: `likes`

```
{
  "id": "string"   // product slug, matches Product.id
  "likes": number  // monotonically increasing counter
}
```

Pydantic `Product` shape (frontend & API)

```
{
  id:             string,        // slug derived from Name
  name:           string,
  category:       string,
  price:          string,
  link:           string,
  media_urls:     string[],
  description:    string,
  store_name:     string,
  city:           string,
  date_acquired:  string,
  featured:       boolean,
  tags:           string[],
  likes:          number          // merged from Mongo at read-time
}
```

7. Frontend Component Reference

`App.js` **— root**

- Sets up React Router with `/`, `/my-shelf` (alias `/my-list`), `/about`.
- Loads products once on mount; tracks `activeId` for the modal.
- Owns the **scroll-lock** via `savedScrollRef` + `useEffect` on `!!activeId`.
- Owns the `likedSet` (Set of liked IDs from `localStorage[sl_liked_v1]`).

`components/ProductDetail.jsx`

The most intricate component. Two layouts coexist in the DOM:

- **Mobile** (`md:hidden`): full-screen, sticky close button at `fixed top-3 right-3 z-30`, scrollable content area, sticky bottom action bar (Add to Shelf + open-link arrow). Gallery has `onTouchStart` / `onTouchEnd` handlers — horizontal swipe >40px advances/rewinds.
- **Desktop** (`hidden md:grid`): 7/5 column grid, fixed top (category + close), title block, scrollable description, fixed footer (meta, tags, actions, prev/next chevrons).

Dialog uses `modal={false}` so Radix doesn't run its own scroll-lock; we render our own backdrop at z-99 (`data-testid="modal-backdrop"`) above the header (z-70) but below the dialog content (z-101).

`components/Header.jsx`

- `fixed top-0 z-[70]`. Goes transparent over the Hero on Home, opaque otherwise.
- Mobile hamburger opens a full-screen `z-65` menu panel.
- Brand mark mixes Bricolage Grotesque + EB Garamond italic ampersand.

`pages/MyList.jsx` **(route** `/my-shelf`**)**

- Filters all products by `likedSet`.
- Empty state: inline `EmptyShelfDog` SVG (line-art bookshelf + confused terrier with question mark) + CTA *Add to Shelf* linking to `/#archive`.

8. Design System

Typography

Token	Family	Use
`font-display`	EB Garamond (fallback Garamond, Times)	Editorial titles, hero, product names.
`font-modern`	Bricolage Grotesque (fallback Manrope)	Brand wordmark.
`font-instr-sans`	Instrument Sans (fallback Manrope)	UI labels, buttons.
body	Manrope	All running body text.
`meta-label`	Manrope tracking-widest uppercase 11px	Editorial meta strips: "[002] / FAVOURITES".

Colour

- Background: `#F7F7F7` (off-white)
- Ink: `#0A0A0A` (near-black)
- Accent dark card (Bento): `#0A0A0A` with white type
- Hairlines: `black/[0.06]` to `black/[0.10]`

Motion

- Card hover: image `scale(1.04)` over 900ms, opacity reveal of arrow chip.
- Modal open/close: backdrop `fade-in-0`, content scaled-in via Radix (200ms).
- Hero blur: opacity from 0 to 1 + slight scale on mount.
- Mobile menu: stagger items by 60ms each.

9. Known Quirks & Hard-Won Lessons

> **Modal scroll-lock** — the single most fragile thing in this codebase. Several iterations failed before this final pattern landed. *Do not refactor casually.*

The pattern (in `App.js`)

```
// Capture BEFORE setActiveId — Radix mount auto-scrolls so reading
// window.scrollY inside the useEffect is too late.
const savedScrollRef = useRef(0);
const openProduct = (p) => {
  savedScrollRef.current = window.scrollY;
  setActiveId(p.id);
};

// Effect depends on !!activeId so prev/next switches don't unlock.
useEffect(() => {
  if (!isOpen) return;
  const savedY = savedScrollRef.current;
  body.style.position = "fixed";
  body.style.top = `-${savedY}px`;
  body.style.left = "0"; body.style.right = "0"; body.style.width = "100%";
  return () => {
    body.style.position = ""; body.style.top = "";
    body.style.left = ""; body.style.right = ""; body.style.width = "";
    // OVERRIDE scroll-behavior:smooth from index.css to avoid 600ms anim.
    html.style.scrollBehavior = "auto";
    window.scrollTo(0, savedY);
    html.style.scrollBehavior = "";
  };
}, [isOpen]);
```

Why each detail matters

- **Capture in click handler, not in effect** — Radix Dialog's mount triggers an auto-scroll up to thousands of px. Reading `window.scrollY` inside the effect captures the wrong value. Iteration 3 of testing found this clearly.
- `position:fixed` **with** `top:-savedY` — only pattern that truly freezes the visual position regardless of focus events.
- **Override** `scroll-behavior:smooth` on cleanup — index.css line 40 sets this globally, which would animate the post-close `scrollTo` over ~600ms. Override to `auto` for an instant snap.
- `modal={false}` on Radix Dialog — using `modal=true` activated `react-remove-scroll` which conflicted with our custom lock and auto-scrolled the page. We render a custom backdrop instead.
- **Custom backdrop at z-99** — the header is z-70, dialog content is z-101. Without a backdrop covering z-71→100, the header is visible above the modal. The custom backdrop hides it.

Other gotchas

- `html, body { min-height: 100% }` in index.css — was previously `height: 100%`, which caused scrollY to reset when `react-remove-scroll` set `overflow:hidden`. Switched to `min-height`.
- Header watches `scrollY > 40` to switch from transparent → opaque. While the modal is locked, scrollY is 0, so the header would normally go transparent. We don't fix this because the backdrop covers it anyway. If you ever drop the backdrop, you'll need to force the "scrolled" state during modal open.
- `Header.jsx` uses `document.body.style.overflow = 'hidden'` for its own mobile menu lock. This is fine on its own but can clash with the modal lock if both opened simultaneously — they don't, but worth knowing.
- Both desktop and mobile modal trees are in the DOM (toggled by `md:hidden` CSS). Test IDs like `btn-interested` exist in the mobile tree, `btn-interested-desktop` in the desktop tree. When writing tests, query

by visibility or by viewport-specific ID.

10. Local Development & Deployment

Frontend

```
cd /app/frontend
yarn install
yarn start         # http://localhost:3000
```

Backend

```
cd /app/backend
pip install -r requirements.txt
uvicorn server:app --host 0.0.0.0 --port 8001 --reload
```

MongoDB

Local instance, no auth. Connection via `MONGO_URL`. The `likes` collection is auto-created on first POST.

Required env vars

File	Var	Purpose
backend/.env	`MONGO_URL`	Mongo connection string
backend/.env	`DB_NAME`	Mongo database name
backend/.env	`GOOGLE_SHEET_ID`	The sheet ID (omit to fall back to the bundled 10 samples)
backend/.env	`GOOGLE_SHEET_GID`	Default `0`
backend/.env	`SHEET_CACHE_TTL`	Seconds, default `300`
backend/.env	`CORS_ORIGINS`	Comma-separated allow-list
frontend/.env	`REACT_APP_BACKEND_URL`	Base URL of the backend (production preview URL)

Deployment

Currently runs on the Emergent platform managed by `supervisor`. The frontend is built with `yarn build` and served as static assets; the backend is run via `uvicorn`. Any container platform (Render, Railway, Fly, Vercel + Render) works as long as the two services can talk to each other and the frontend is given the backend URL via `REACT_APP_BACKEND_URL`.

11. Prioritised Backlog

P1 — High value, low effort

- Header "Refresh" pill that calls `GET /api/refresh` and shows a tiny toast.
- Per-product Open Graph + Twitter card meta tags pre-rendered server-side. Requires a small SSR shim or static generation step.
- Share button per product (copy URL with anchor / Web Share API on mobile).

P2 — Delight

- Lenis momentum scrolling.
- View Transitions API for card → modal shared-element animation.
- Keyboard arrow keys inside the gallery (currently only switches products).
- "Liked by you" sub-shelf at top of archive when `likedSet.size >= 1`.

P3 — Growth

- Email capture for "tell me when new things are added" — would require a backend list + a transactional email integration.
- Server-side image proxy + responsive `srcset` resizing for ultra-fast loads on slow networks.
- Shareable "My Shelf" — encode the liked-IDs into a URL hash (or a tiny share token) so visitors can DM their curated mini-shelf to friends.

12. Conversation Timeline

This is a chronological summary of the build, distilled from multiple sessions. Each phase ends with a check-in / verification gate.

Phase 1 — Foundations

- FastAPI + React + MongoDB scaffolded.
- Google Sheet ingestion with tolerant CSV parser, header normalisation, 5-min cache, and 10-product sample fallback.
- Likes endpoint backed by MongoDB.

Phase 2 — Editorial UI

- Cinematic Hero with EB Garamond, blurred lifestyle background, brand wordmark.
- Bento Grid with 4 featured + a black accent card.
- Archive section with sticky category filter pills.
- Product Detail modal with media gallery, story, "I'm interested" button.
- Routing: Home, My List, About.

Phase 3 — Mobile polish

- 2-column bento on mobile, hero text centring, hamburger menu, sticky filters fix.
- Resolved 10 mobile UI bugs across overflow, z-index, typography, thumbnail outlines.

Phase 4 — Modal scroll-lock saga

This took several iterations and is the most subtle work in the codebase. Key data points captured by the testing agent across iterations:

- *Iter 1:* opening the modal jumped the background to Hero.

- *Iter 2:* initial fix scrolled-to-top because `html { height:100% }` + `react-remove-scroll` caused scrollY to reset.
- *Iter 3:* capturing scrollY inside the `useEffect` read the wrong value (Radix mount auto-scrolled before the effect ran).
- *Iter 4:* Final fix — capture in the click handler via a `useRef`, plus override `scroll-behavior:smooth` on cleanup. **100% pass.**

Phase 5 — Final UX pass (current)

- Modal layered properly with a custom z-99 backdrop covering the navigation.
- Mobile modal redesigned: scrollable content, sticky close at top-right, sticky bottom action bar, swipe-to-change image.
- Renamed *I'm interested → Add to Shelf*; toggled to *On your shelf* after liking.
- Renamed *My list → My shelf*, route `/my-shelf`, with a custom inline-SVG empty state (line-art bookshelf + confused dog).
- Bonus: fixed `fetchpriority` → `fetchPriority` React warning in `Hero.jsx`.

13. Appendix

Test reports

The most recent automated test run (`/app/test_reports/iteration_4.json`) reports **100% pass** on critical assertions:

- Desktop & mobile scroll preservation: pre = post (1500 → 1500, 600 → 600).
- Backdrop covers entire viewport including the navigation header.
- "Add to Shelf" → "On your shelf" toggle correctly increments likes.
- "My shelf" nav label, `/my-shelf` route, empty-state SVG + CTA.

data-testid registry

Test ID	Where
`site-header`	Header
`brand-link`	Header brand
`nav-home / nav-myshelf / nav-about`	Header nav
`mobile-menu-toggle / mobile-menu-panel`	Mobile menu
`archive-section`	Archive
`product-card-{id}`	Each card in the grid
`card-likes-{id}`	Likes badge on card
`product-detail`	Modal root
`modal-backdrop`	Custom backdrop (z-99)

`close-detail / close-detail-mobile`	Desktop / mobile close buttons
`btn-interested / btn-interested-desktop`	Add to Shelf (mobile / desktop)
`btn-source / btn-source-desktop`	Open-link arrow
`product-prev / product-next`	Modal navigation chevrons (desktop)
`product-prev-mobile / product-next-mobile`	Modal navigation (mobile)
`thumb-{i}`	Desktop gallery thumbnails
`mobile-gallery-dots`	Mobile gallery pagination dots
`myshelf-page / myshelf-empty / myshelf-cta`	My Shelf page

localStorage keys

- `sl_liked_v1` — JSON-stringified array of liked product IDs.

End of handoff document. Combine with the GitHub repository pushed via the platform's *Save to GitHub* action and the team has everything they need to continue.